

ALAN MATTHEW

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EDUCATION

Yale–NUS College

Aug 2020 – May 2025

Bachelor of Science (Honours) in Math, Computational and Statistical Sciences | GPA: 4.87/5.00

Coursework: Asset Pricing, Statistical Inference, Stochastic Modelling, Machine Learning

Activities: Teaching Assistant for Data Structures & Algorithms, Probability Theory, Statistical Inference

Awards: Kewalram Chanrai Full Scholarship

UWC Li Po Chun

Aug 2018 – May 2020

International Baccalaureate Diploma (40/45)

Coursework: HL: Math, Physics, Chemistry (7/7/7)

Activities: Working with Domestic Helper Club (US\$5,000 & 250 books for Wonosobo library); United World Schools (US\$16,500 for schools in Yern Jas); Dragon Dance

Awards: UWC Scholarship

EXPERIENCE

DRW

HFT Stock Options Trading

Quantitative Trading Analyst

Jan 2026

- Conducted high-frequency signal research, analyzing market microstructure and large-scale trading data to identify and evaluate systematic trading opportunities.
- Developed and refined volatility and interest rate models to support options pricing, relative-value analysis, and trading decisions.
- Actively traded options and managed live systematic strategies, monitoring execution, portfolio exposures, and real-time risk while adjusting hedges and positioning as market conditions evolved.

Caladan (AlphaLab Capital)

Cryptocurrency Market Making

Software Engineer

Jul 2025 - Oct 2025

- Designed and built a generalized caching layer in Python to manage storage and fast retrieval of large-scale time-series data across S3, SQL, etc., reducing query latency and improving pipeline throughput by 97%.
- Reimplemented ETL job services with fault-tolerance and async design (Python event loops, DAG modeling), improving recovery, consistency across dependent tasks, overall maintainability, and clarity of error reporting.
- Built a fault-tolerant 24/7 ETL service for real-time PnL computation, enforcing invariants for correctness and exposing statistics for business analysis, with infrastructure libraries supporting QuestDB and Grafana logs.
- Developed and maintained ETL pipelines, automating post-trade and transaction cost analysis (TCA) across high-frequency trading algorithms, delivering insights on hedge timing, execution quality, and price markouts.
- Designed and implemented robust NAV-based PnL attribution frameworks, producing granular breakdowns of performance by hedge leg, execution venue, etc., for comprehensive session-level trading reviews.

Caladan (AlphaLab Capital)

Cryptocurrency Market Making

Software Engineering Intern

May 2024 - Aug 2024

- Implemented cold and hot path components for C# trade servers, expanding crypto trading to new exchanges.
- Developed data ETL pipeline visibility service in Python; Rewrote core services using async to improve speed.
- Resolved critical bugs in algorithm trading logic and fixed out of memory issues, saving monthly costs by 80%.
- Expanded Python analysis libraries for deeper insights into market volume, PnL, and trading statistics.

Evooq

Fintech SaaS

Software Engineering Intern

Jan 2024 - Apr 2024

- Created a Python LLM chatbot that translated natural language queries into SQL for a MariaDB database.

- Created a Python tool to generate trading ideas for a given portfolio with specific goals using RAG.

Switchero Labs

Cryptocurrency Exchange

Software Engineering Intern

Oct 2022 - Dec 2022

- Leveraged Golang to automatically link markets and liquidity pools, eliminating manual work and securely allocating USD \$130+ million of capital across the exchange.
- Implemented a microservice in TypeScript to monitor and collect data on CDP transactions and visualize trading activities using React, serving 35,000 daily visitors.
- Created a Golang API to aggregate and consolidate data from multiple databases into a single, unified API call, simplifying the information search process for 96,000+ users.

Investree

P2P Fintech Lending

Software Engineering Intern

Feb 2022 - Aug 2022

- Implemented new features using Vue.js and Kotlin to improve the presentation of financial statements and key metrics, boosting user reviews by 17%.
- Designed and developed a covenant page from scratch using Vue.js and Kotlin, simplifying the contractual agreement process for over 59,000 business loans.
- Developed and deployed RESTful APIs in Kotlin for data querying, processing, and validation from Postgres databases, serving the frontend needs of over 100,000 users.

Singular Asset Management

Asset Management

Equity Trading Intern

Sep 2021 - Jan 2022

- Traded Indonesian equities on IDX, supporting a long-only portfolio with PM; generated 11% returns.
- Performed in-depth company analysis, industry research, competitor analysis, ESG analysis, and on-the-ground research to identify key trends across diverse sectors in Indonesia.
- Liaised with sell-side analysts and industry experts during broker calls and investor conferences to refine market perspectives and uncover new trading opportunities.

LEADERSHIP

Perception Gap

International Network

Founder

Jul 2022 - Jul 2023

- Started and manage an international equity research club that grew from 4 friends to a global network of 17 members at institutions like Cornell, LSE, UPenn, Wellesley, HKU, and HKUST.
- Taught accounting and financial modelling to members, leading them to write comprehensive equity research reports on Tesla, Boeing, Lockheed Martin, Grab, and more.
- Organized mock interviews and quiz nights so members can network, sharpen business knowledge, and share market happenings—helping third-year members land investment banking and private equity internships.

OPEN SOURCE

panglesd/diffcensible

Accessible Diff Viewer (OCaml)

- Developed side-by-side diff rendering with numbered lines, color highlighting, etc., to enhance user experience.
- Implemented data structures (e.g., zipper, block-based storage) and linear-space diff algorithms to enable precise word-level comparisons across multi-line changes and improve performance in large-scale file navigation.

PROJECTS

- **yokurang/poe2-arbitrage**: Built an ML pipeline to scrape item prices and attributes and estimate FX rates via VWAP. Developed regression and closed-form models to estimate attribute contributions to item value.
- **yokurang/mytree**: A published Rust tool for displaying directory trees with custom filters. 2k+ downloads.

SKILLS

Languages: Python, Rust, C++, OCaml, C#, TypeScript, JavaScript, SQL, Kotlin, Scala, Golang, Coq

Frameworks: React, Next.js, Flask, PyTorch, TensorFlow, scikit-learn, AWS (Lambda, DynamoDB)